# The European Union: Institutions, Policies and Future Challenges.

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**Manfred Grasserbauer** *

**Technische Universität Wien**

* **Director at Joint Research Centre, European Commission**
**1997 – 2007**

# The European Union:
## Institutions, Policies and Future Challenges.


Ch. 1: The History of Europe.
Ch. 2: The Evolution of the European Union.
Ch. 3: Institutions and Governance of the European Union.
Ch. 4: The Budget of the European Union.
Ch. 5: Major Policies and Activity Areas of the European Union:
- Ch. 5.1: The Single Market.
- Ch. 5.2: The Solidarity Policies.
- Ch. 5.3: The Innovation Policies.
- Ch. 5.4: The Economic and Monetary Union.
- Ch. 5.5: Home Affairs: Freedom, Security and Justice.
- Ch. 5.6: Foreign Affairs: External Security, Trade and Development Cooperation.


*Information Sources: If not otherwise indicated European Commission and other European Institutions, Wikipedia.*

Recommended Literature:
Gabriel Felbermayr “Europa muss sich rechnen.”, Verlag Brandstätter 2023

# Ch. 1: The History of Europe:

## The Origin of Europe.

**Greek mythology:**

**Europa was the daughter of the Phonecian king Agenor from Tyros and for whom the continent Europe was named.**

Zeus abducted her in the form of a white bull from Tyros to Crete where their son Minos was born.

Minos founded the first European empire.

Europa and Zeus as a bull on a Greek vase.
Tarquinia Museum, circa 480 BC.

Europe was created in violence by the supreme god Zeus:
Condemned to permanent conflicts between its peoples,
but also blessed to create the greatest achievements of mankind.

# First European Civilisation:
## Minoan Culture Crete 3.000 - 1.000 BC

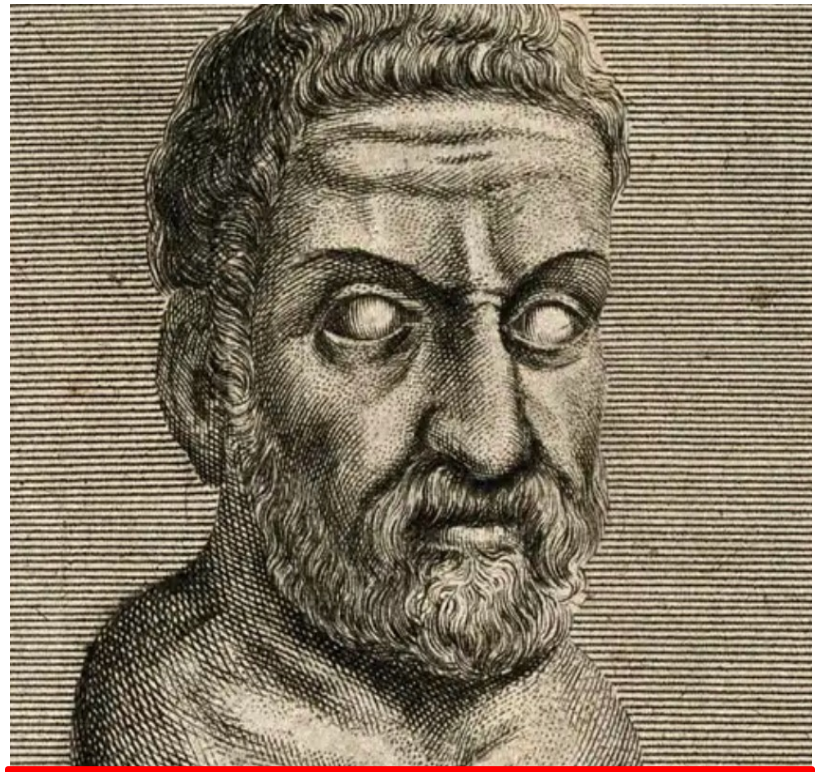
**Disk of Phaistos:**

First „written document“.

Disk of fired clay from the palace of Phaistos in Crete possibly dating to the Bronze Age (1.800 BC).
Symbols printed in soft clay.

# The Foundations of Western Civilization: Major Greek Philosophers.

## Thales (625-546 BC): The First Greek Philosopher.

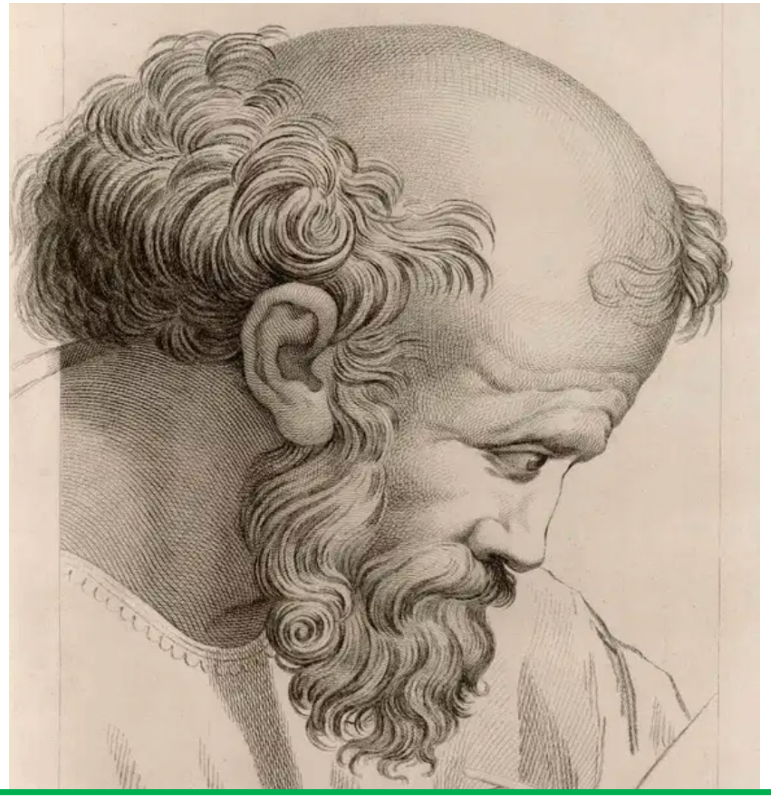


Water is the fundamental substance of the universe.
Introduced the concept of natural explanations.

## Heraclitus (535-475 BC): The Philosopher of Change.

„Panta rhei.“ – Everything is in constant flux.
Logos – There are principles which govern the universe.

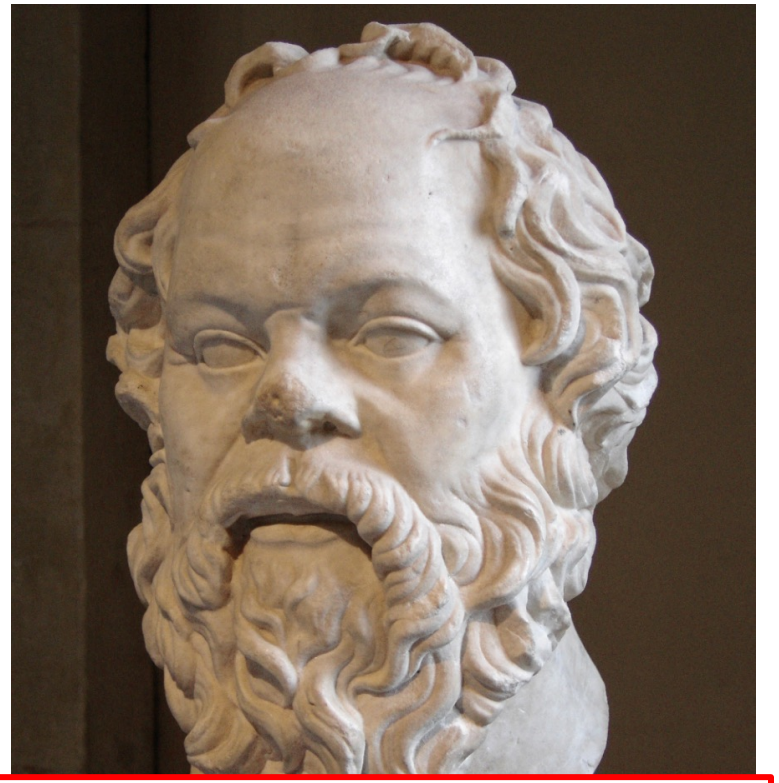
## Pythagoras (570-490 BC): The Father of Mathematics.



Mathematics is the foundation of reality.
Key mathematical concepts.
Concept of deductive reasoning.

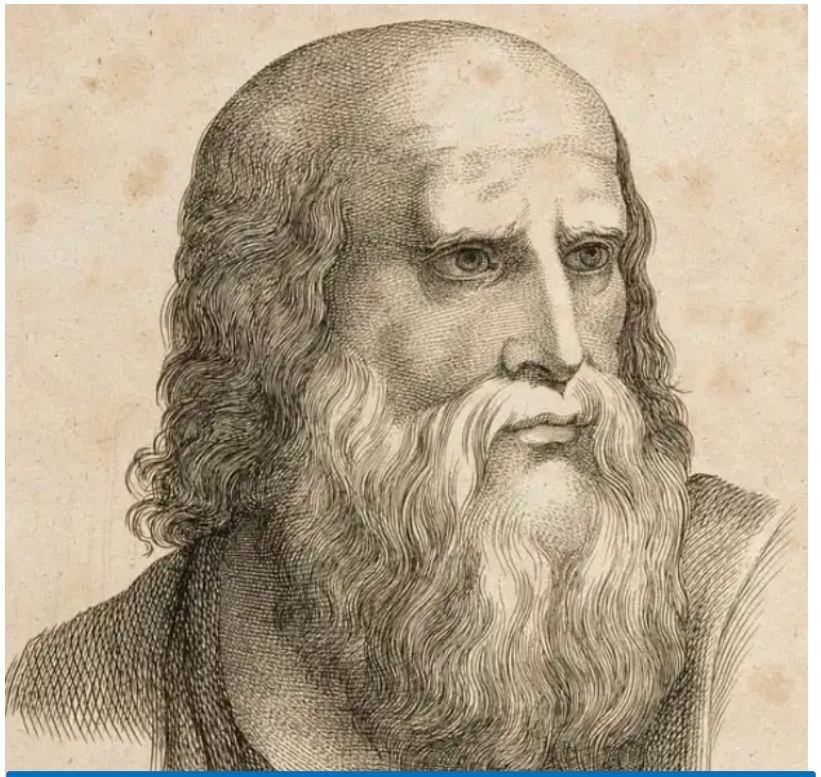
# The Foundations of Western Civilization: Major Greek Philosophers.

## Socrates (469-399 BC): Father of Western Thought.



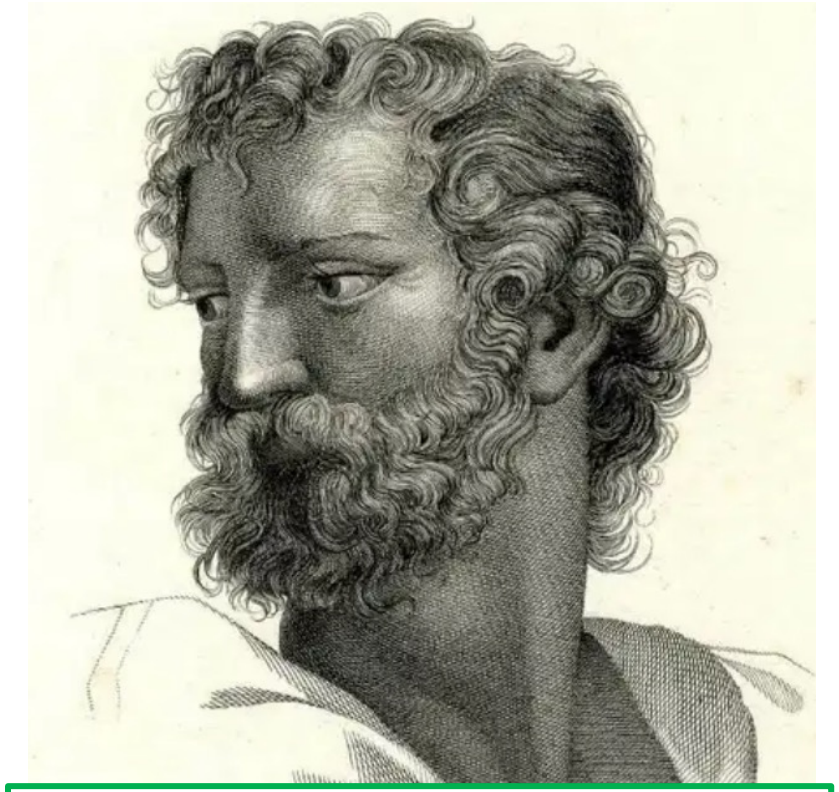
Socratic method of inquiry.
True knowledge is virtue.
Importance of self-reflection and the pursuit of knowledge.

## Plato (427-347 BC): The Most Famous Philosopher.



The Theory of forms.
Limitations of our senses.
Emphasis on reason and the pursuit of knowledge.

## Aristotle (384-322 BC): The Father of Logic.



Empiricism and observation.
Developed the science of logic.
Ethics and the pursuit of happiness.

# The Foundations of Western Civilization:
## The Greek Philosophical Schools.

•Philosophical schools of Greece were centers for creation of knowledge: ethics, concepts of states, principles of law, mathematics, geometry, astronomy…

School of Athens founded by Plato in 385 BC (Raffaello).

But also individualists were popular:
**Diogenes (404-323 BC) - The Cynic.**

„Be free from social conventions and expectations.“

# The Foundations of Western Civilization:
## Governance.

•The governance systems of the Western world were developed in Rome (500 BC till 500 AD).

The Roman Empire 117 AD: 70 million people (20% of the world‘s population) and 50 nations. Autocratic central government with 2 official languages – Latin and Greek.

A Roman aureus struck under Augustus AD 13–14.

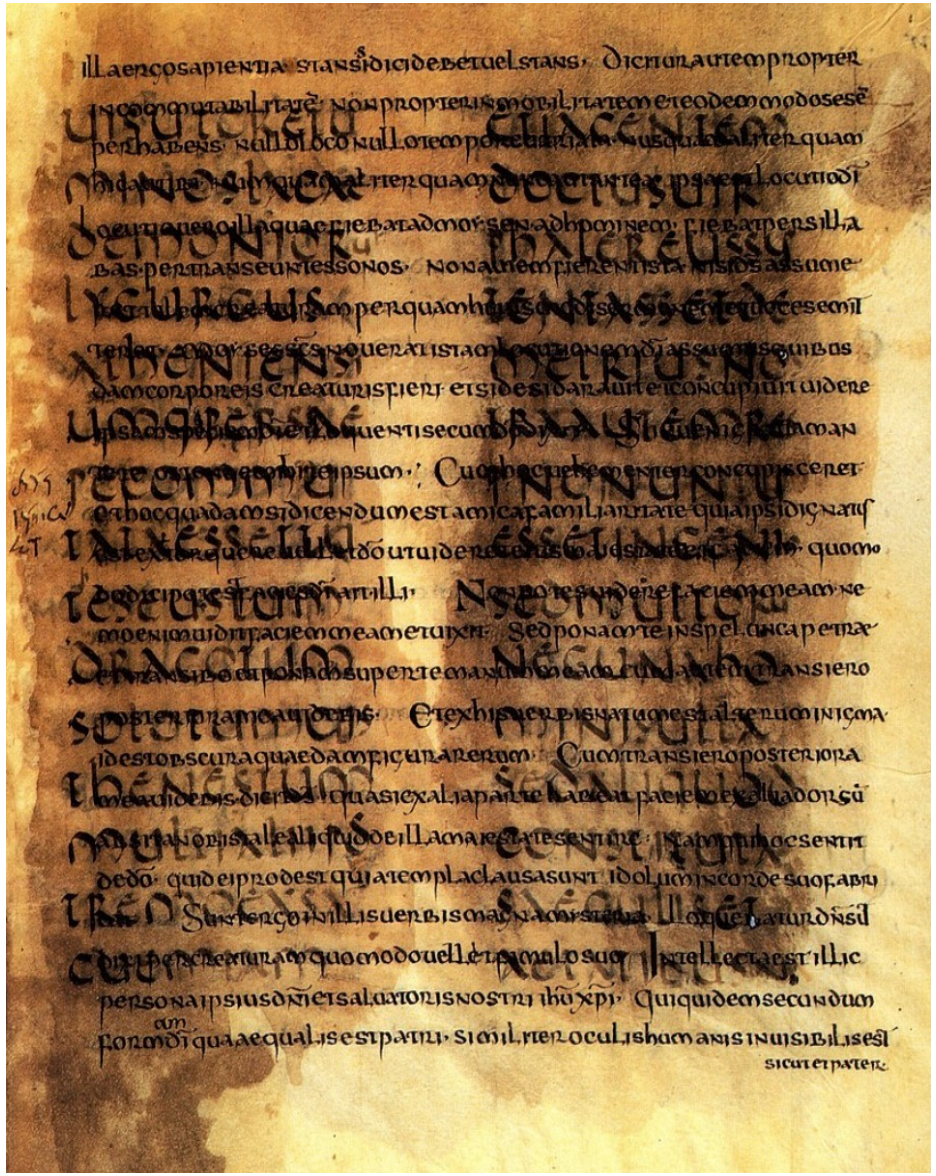
Massive progress in art, engineering, shipping, trade, organisational forms.

Mars and Venus in wall painting in Pompei.

# The Foundations of Western Civilization:
## Roman Law.

**Marcus Tullius Cicero (106-43 BC):**
**Statesman, lawyer, founder of our legal system.**

- **Major works:**
  - De Re Publica
    (About the State)
  - De Legibus
    (About the Laws)
  - Tusculanae Disputiones
    (About Ethics)
  - De Fato
    (About Free Will)
  - De Officiis
    (About the Attitudes
    of Politicians: Prudentia, Iustitia,
    Fortitudo, Temperantia)



*De Re Publica,* Biblioteca Apostolica Vaticana Rome

# The Foundations of Western Civilization:
## From a Polytheistic to a Christian Society.

**Flavius Valerius Constantinus: Roman Emporer 306-337 AD**

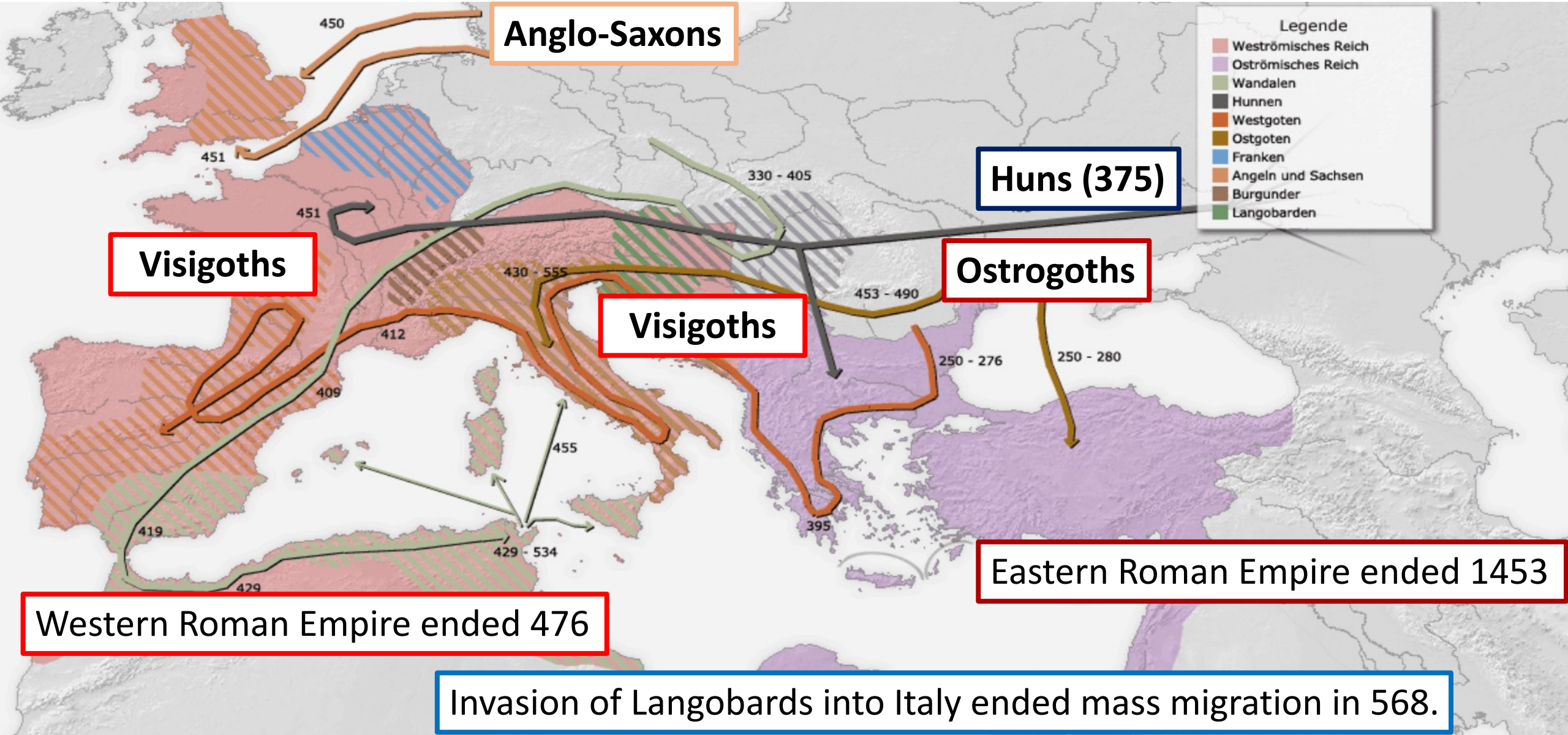
**Edict of Milan 313:**
Christians could execute their religion freely.

Constantinus reaching out to the hand of God.
„In hoc signo vinces.“

**Flavius Theodosius: Roman Emporer 379-395 AD**

**Edict of Thessaloniki 380:**
Catholicism official State religion of the Roman Empire.

# The Collapse of the Roman Empire:
## Mass Migration and Invasion of Tribes from North and East (375-568).




Western Roman Empire ended 476

Eastern Roman Empire ended 1453

Invasion of Langobards into Italy ended mass migration in 568.

# The Growing of a Frankonian Empire (481-814):
## A New Center of Europe.

Bust of Charlemagne
Aachen Cathedral
Treasury

**Charles the Great**
(*Carolus Magnus, Karl der Große, Charlemagne)*
founded the Frankonian Kingdom.

He was crowned by the Pope as „Kaiser“ of the New Holy Roman Empire in 800 in Rome.

Established a “Christian Europe”.

It encompassed what is today seen as the core of the European Union: Germany and France.

# Europe in the Middle Ages:
## Development of a New Society.

**Saint Francis of Assisi,**
**1181-1226**

Founder of the Franciscan Order:
500 monasteries in Europe.

Flourishing communites and a rich and unique
“cultural landscape” developed.

# The Middle Ages:
## Masterpieces of Art.

Capella degli Scroveni, Padova,
with frescos by Giotto (1267-1337)

Santa Maria Assunta, Siena, 13th/14th century.

# Europe at the Beginning of a New Age (1500):

## A Multinational Diverse Continent.

Map of Europe Jan. 1500

**Imperial reign with absolute powers:**

France (Bourbon Kings)

England (Tudors)

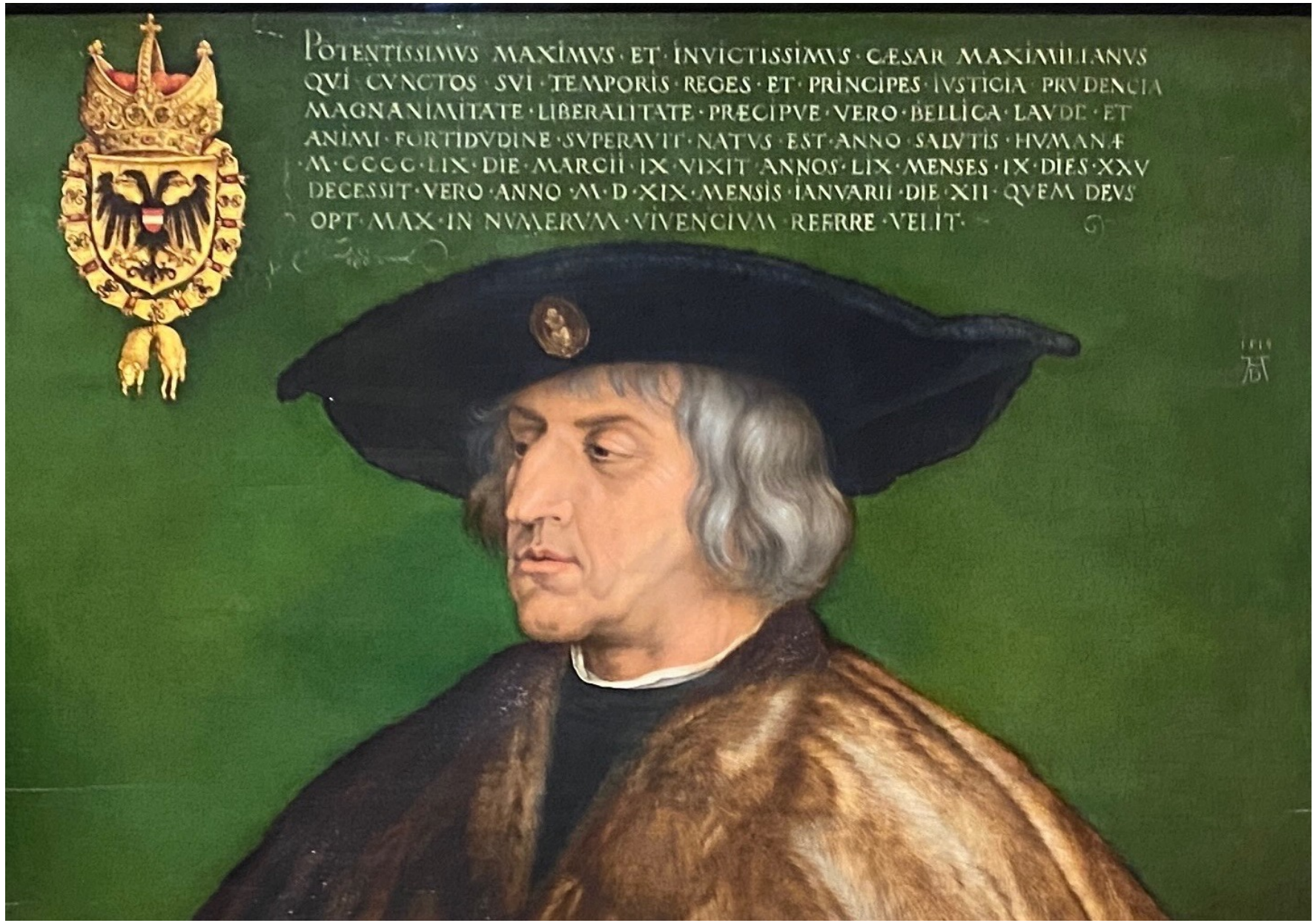
Spain (Habsburg Dynasty)

Germany-Austria (Habsburgs)

Ottoman Empire

Henry VIII – King of England (1509-1547).

# Maximilian I. – Founder of the Habsburg Empire.

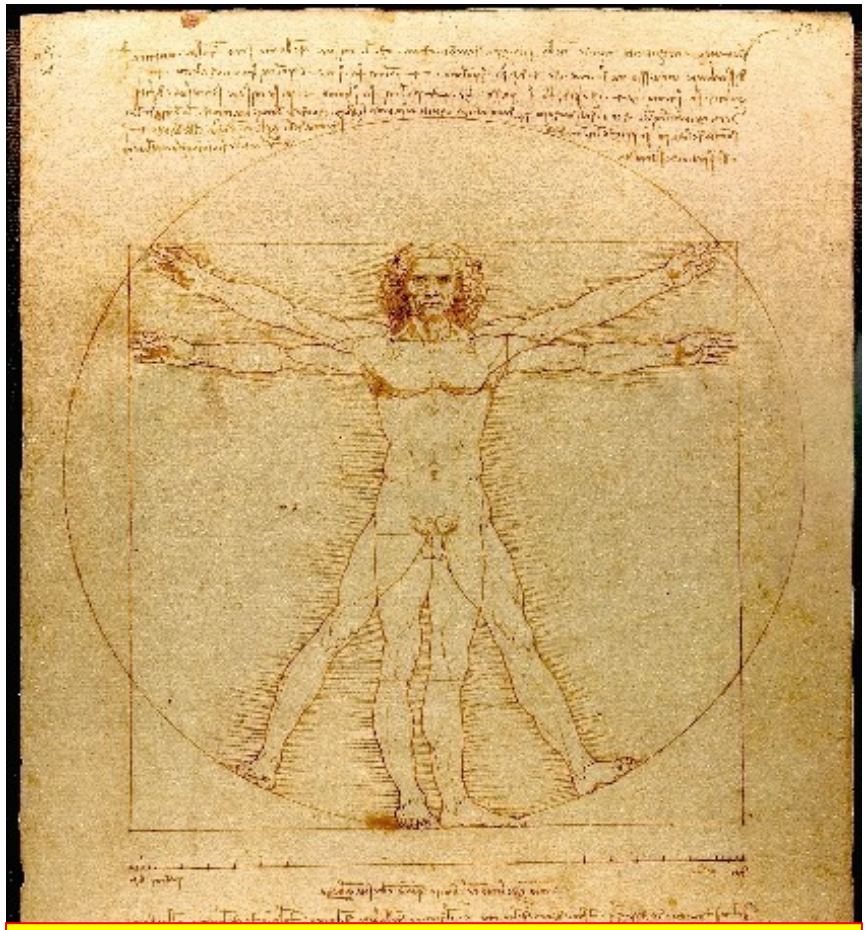



Kaiser of the Holy Roman Empire of German Nation.

Albrecht Dürer (1519).

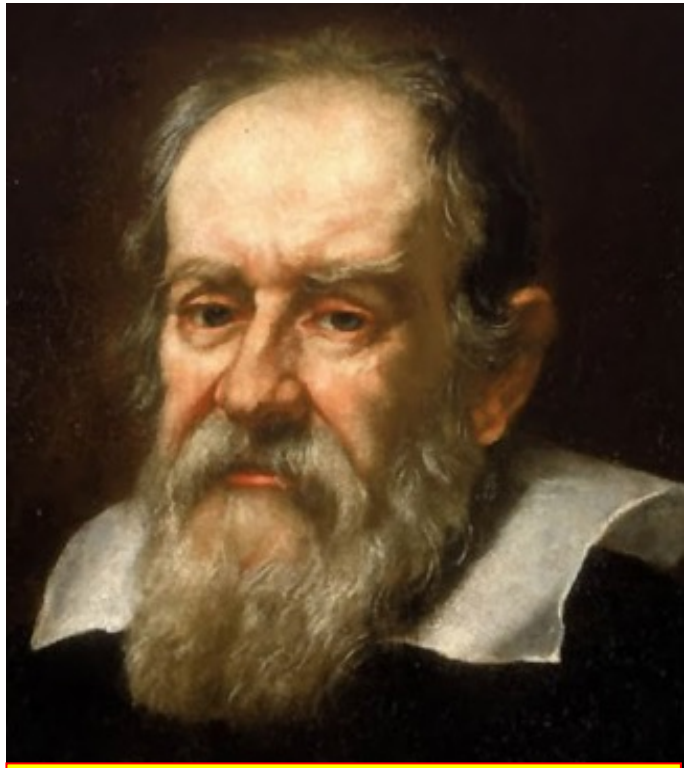
# The Renaissance:

## Rediscovery of Classical Learning in the 14th/15th century.



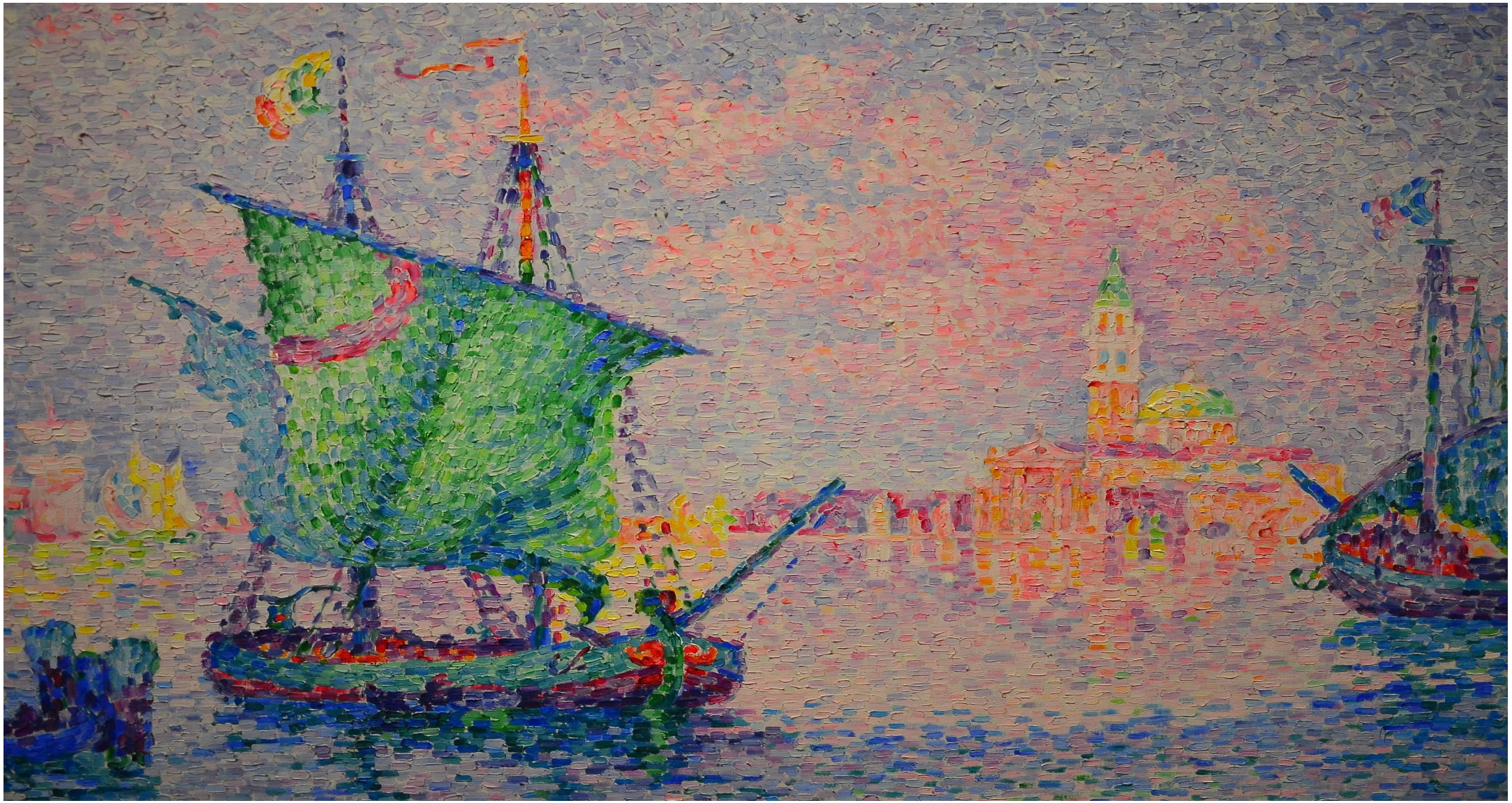
Leonardo da Vinci‘s
Vitruvian Man
of perfect proportions.

- Revolutionary scientific discoveries and inventions would provide the basis for a rapid development of a modern society:
  - Heliocentric system (Galileo Galilei, Nikolaus Kopernikus)
  - Universal Laws of Mechanics (Isaac Newton)
  - Wave theory of light (Christian Huygens)
  - Printing press (Johannes von Gutenberg)
  - Compass (Flavio Goia)
  - Mechanical clock (Peter Henlein)



Galileo Galilei (1564 – 1642)

# The Renaissance:
## Venice - Ruler of the Mediterranean Sea.



Paul Signac: Venice and the Pink Cloud (1909)

# Masterpieces of the Renaissance.

Sandro Botticelli: La Primavera (1482)

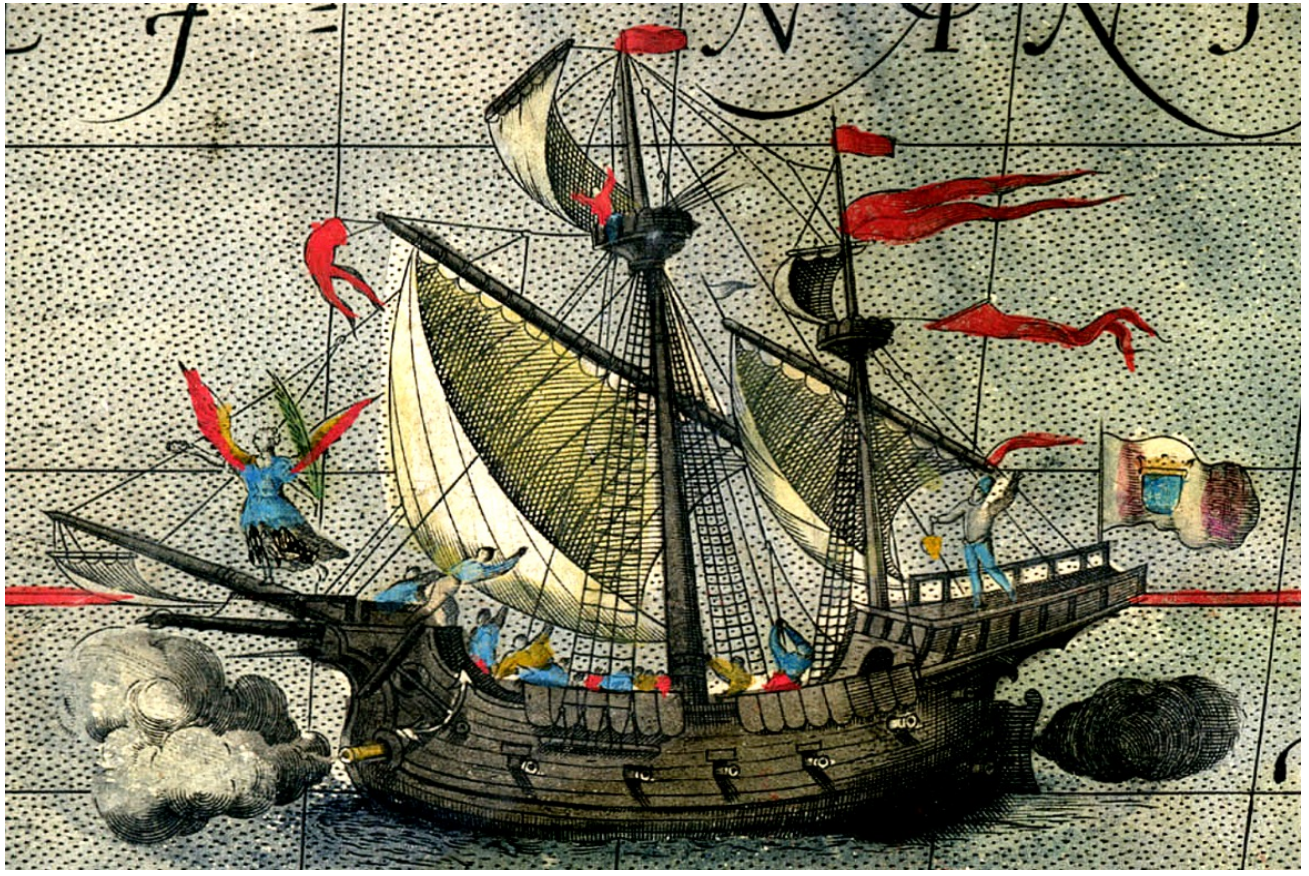
# The Age of Discoveries.

Over the next 300 years, European seafarers set out to explore the unknown world and discover new countries and continents:

America, Africa, India, Southeast Asia, Australia.

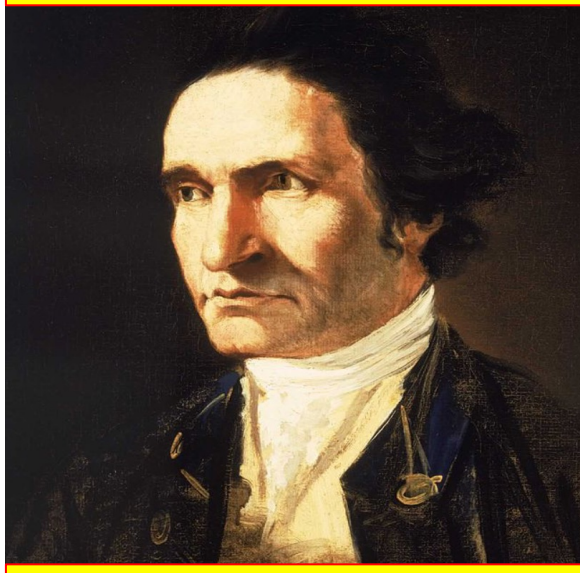
Cristoforo Colombo (1451-1506)

Vasco da Gama (1469-1524)



Victoria – only ship to return from Magalhães expedition.

Fernão de Magalhães (1480-1521)



James Cook (1728-1779)

# Colonization.

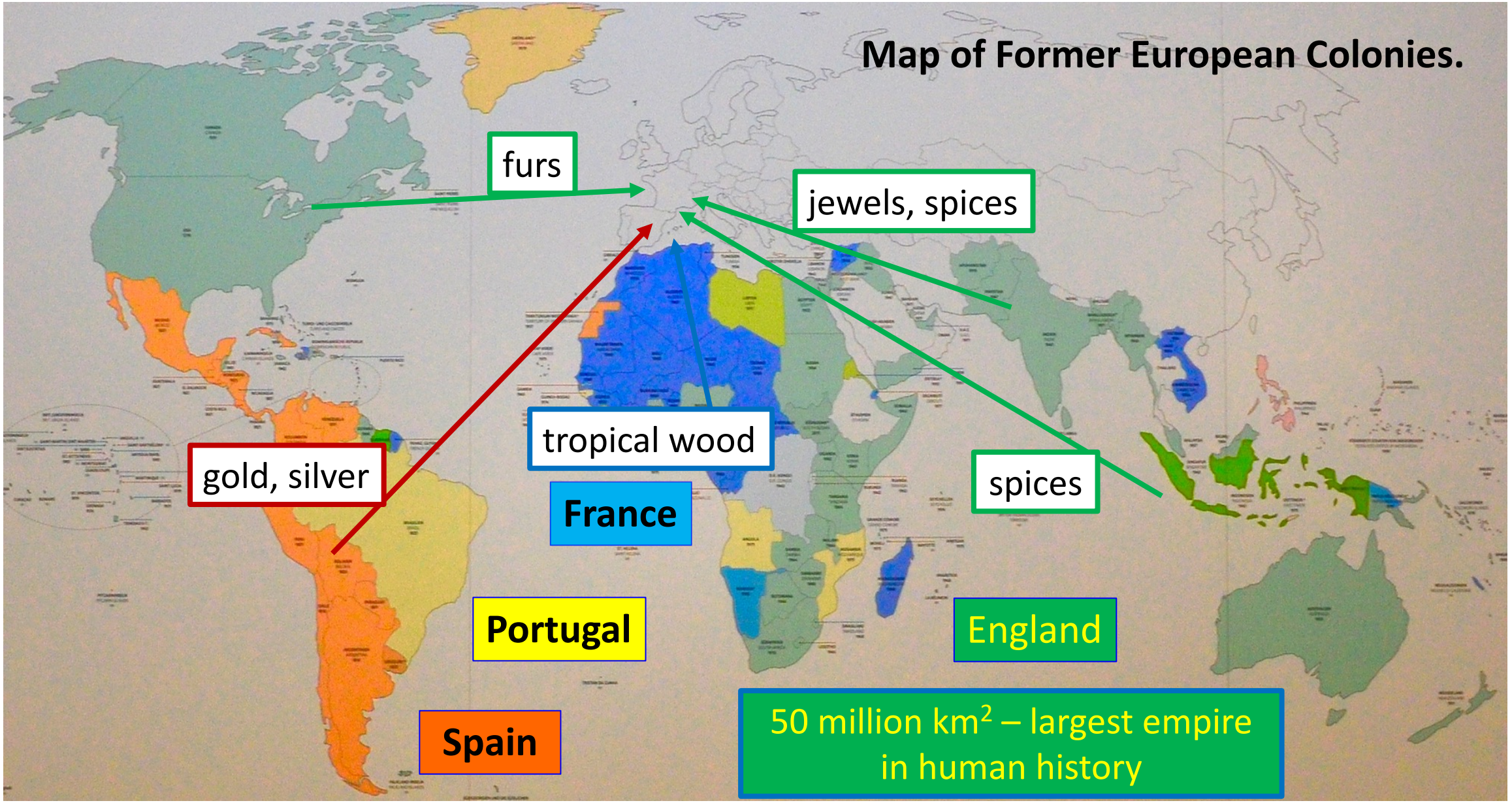
- Exploration and imperialistic conquest brought much of the Americas, Asia, and Africa under European control.
- First colony: the Aztec empire became New Spain.

Quetzal feathered headdress, Mexico (Weltmuseum Wien)

Hernando Cortes (1485-1547)

**Conquest of the Aztec Empire:**
500 soldiers with horses and firearms defeated an Aztec army of more than 50.000 with the help of an uprising of other peoples suppressed by the Aztecs.

# „Masters of the Universe“.



# Golden Altar of Santa Maria de la Sede, Cathedral of Seville.

2.500 kg gold used for plating the figures.

# „Britania rule the waves!“

- **Huge and powerful trading companies were formed for the exploitation of the new colonies.**
- **East India Company** (1600–1874) was an English enterprise formed to trade in the Indian Ocean from India to Southeast Asia.

MS Cutty Sark

- 200 ships.
- 50.000 employees.
- Half of the world‘s trade 1750-1850:
  - Mainly cotton, silk, indigo dye, sugar, species, tea, opium.
- *De facto* rulers of the Indian subcontinent till 1857.

Company flag (1801)

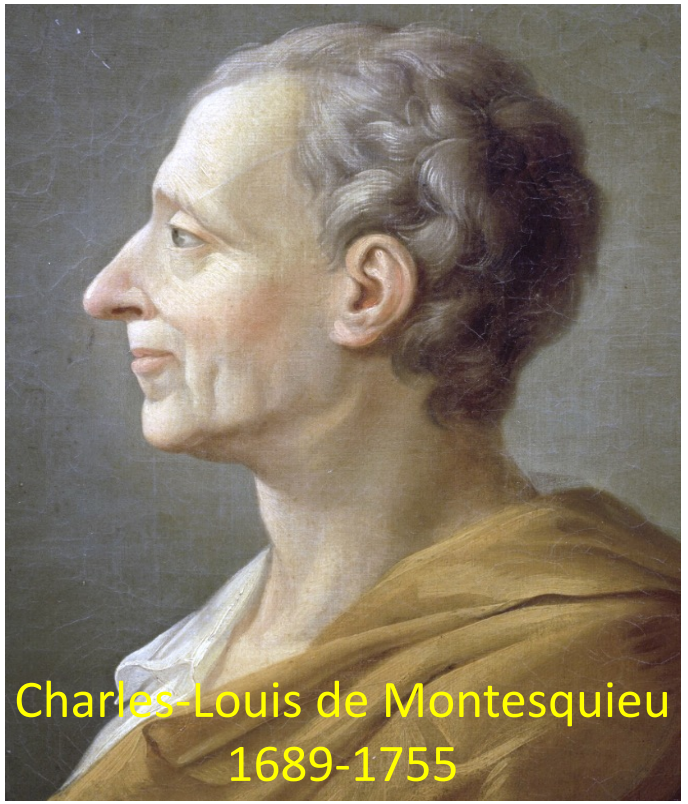
Coat of arms (1698)

# The Age of Enlightenment.

Immanuel Kant
1724-1804

**Charles-Louis Baron de Montesquieu:**
*De l'Esprit de Loi* (1748) –
Principle of the separation of powers
(legislation, executive, jurisdiction).

**Immanuel Kant:** „*Sapere Audere*“ (1784) -
“Dare to know, Dare to think for yourself.”

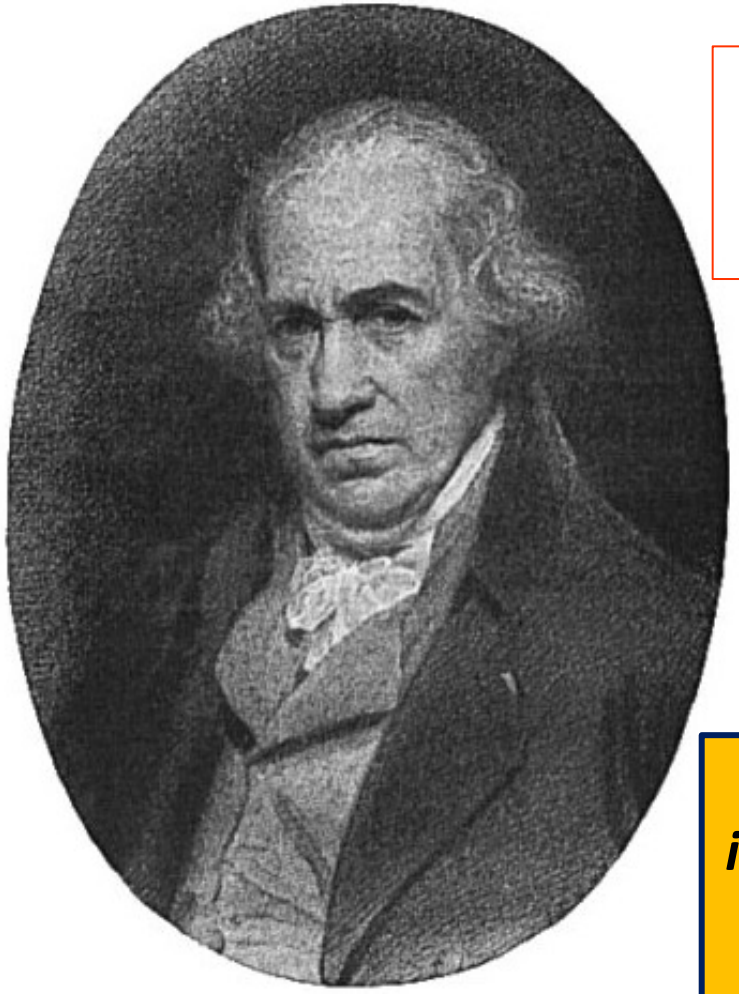

Charles-Louis de Montesquieu
1689-1755

**French Revolution 1789:**
**Liberté, Fraternité, Égalité.**

***Europe is the only region in the world, where such a radical change of thinking occured.***
***Provides the basis of our liberal and free societies.***
***USA implemented these ideas in its constitution 1789.***

# The First Industrial Revolution.

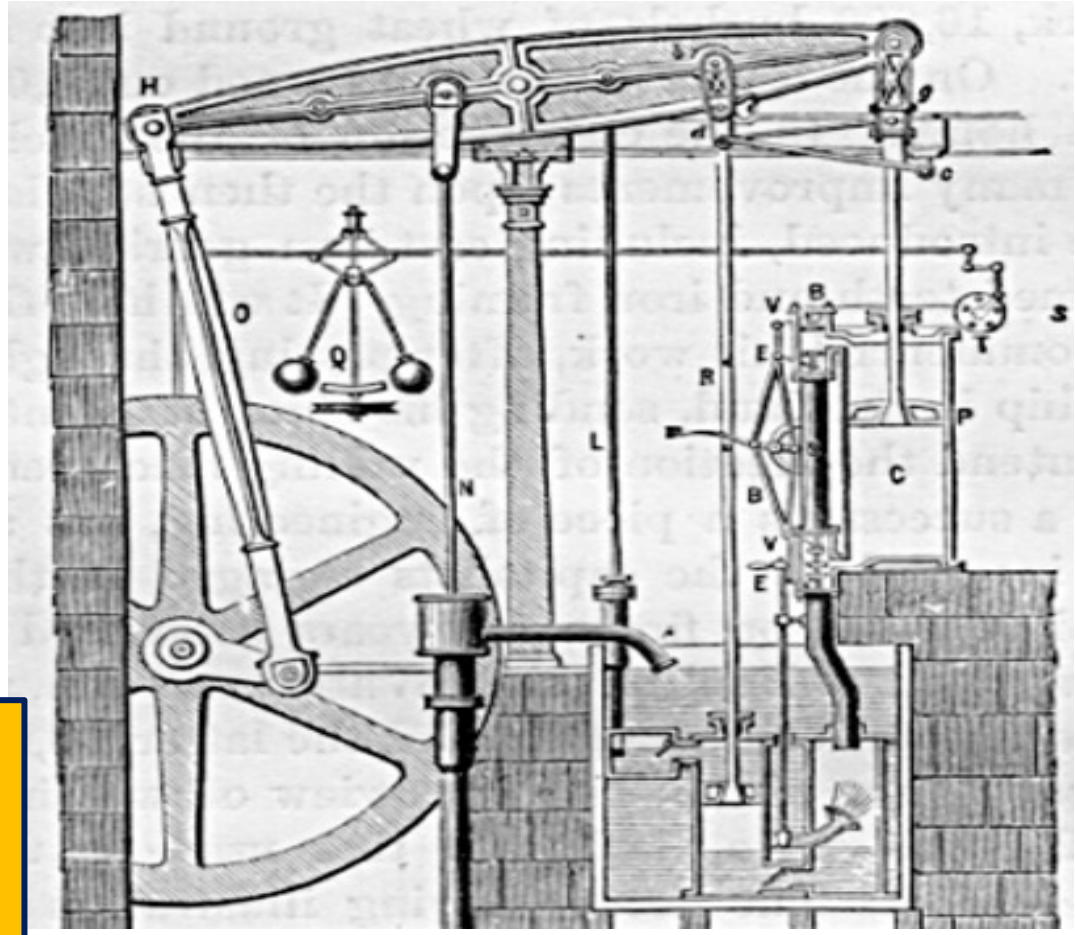
- The First Industrial Revolution from 1750 till 1850 promoted major innovations:
steam engine to introduce mechanical labor, mechanical transport, such as the railway and steam ships.
- Energy produced by burning of carbon to $CO_2$: charcoal, brown and black coal.



Steam engine designed by Boulton & Watt. Drawing from 1784.

**James Watt,**
born Greenock, Scotland 1736, died Handsworth, England 1819

***One of the most important inventions in human history: Replaced human labor by mechanical energy.***



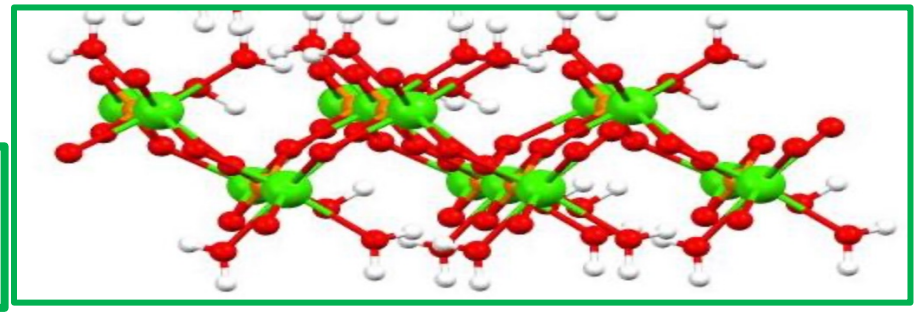
# The Rise of Chemistry.

Antoine Lavoisier (1743 – 1794)

- Development of chemical sciences provided the basis for new materials, new medicines, new forms of energy production, a better understanding of nature and its processes and thus improved food production.

- Discovery of the role of nutrients for plant growth and development of the first chemical fertilizer allowed to combat hunger and starvation in a rapidly growing population.

Justus von Liebig (1803 – 1873)



Superphosphate – the first chemical fertilizer.

# Food Supply in the 19th Century.

Vincent van Gogh “De Aardappeleters“ (1895).

# The Second Industrial Revolution.

•The second industrial revolution from 1850 till 1940 brought about major innovations, like:

European railway network 1850

•modern steel production ➡ railways

•electricity ➡ radio, electric lightening, telephone

•combustion engine ➡ automobiles

•pharmaceuticals ➡ modern medicine

Syphilis caused by bakterium treponema pallidum treated with Arsphenamin (1910).

# The Second Industrial Revolution.

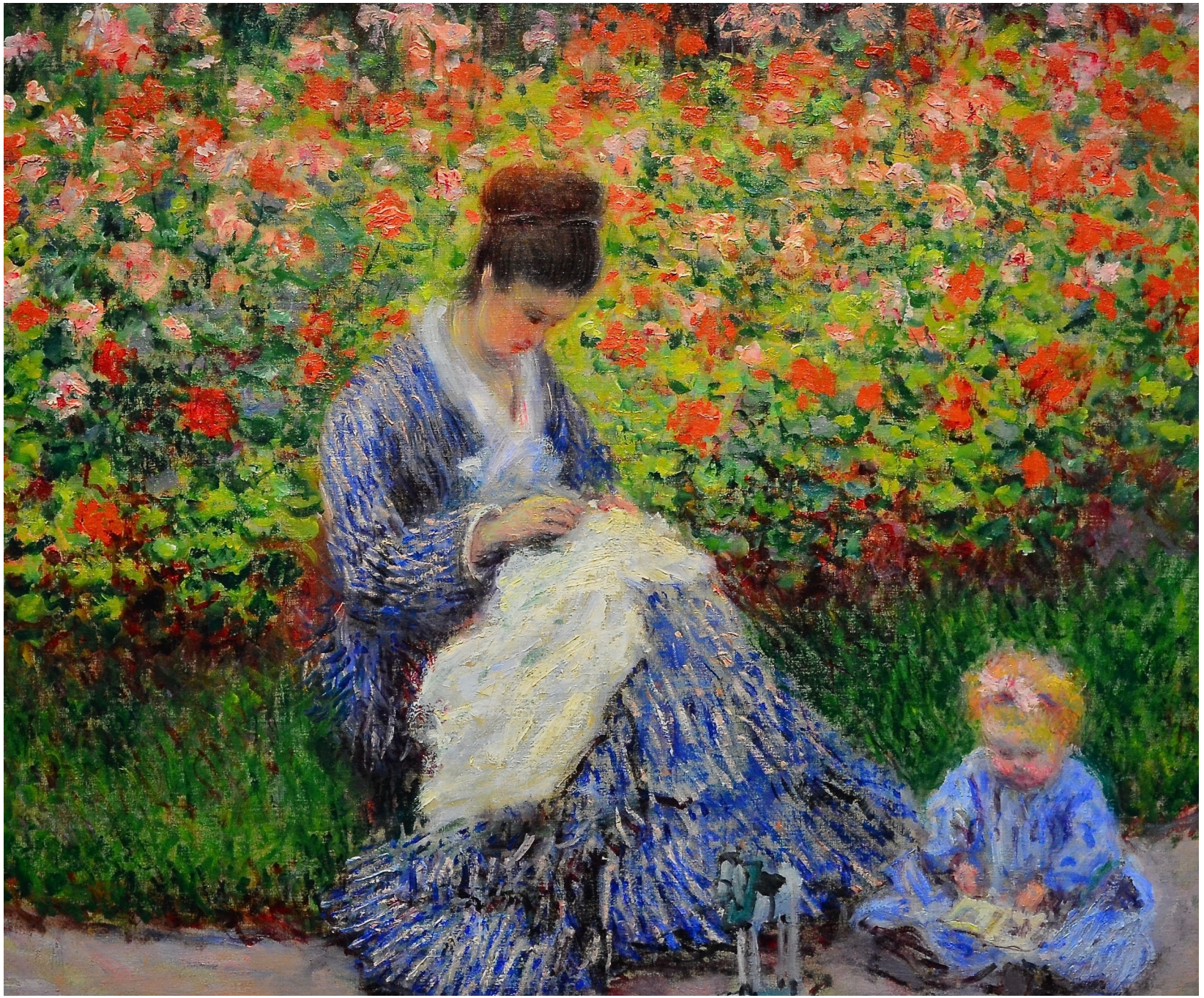
•The second industrial revolution enabled travelling and introduced tourism.

Lionel Feininger “Locomotive” (1910).

# In the 19th Century Europe had become the most glorious part of the world.

London 1851: Crystal Palace 560x137m

Tour Eiffel Paris: Height 330m



Idyllic Life of Upper Middle Classes in Europe.

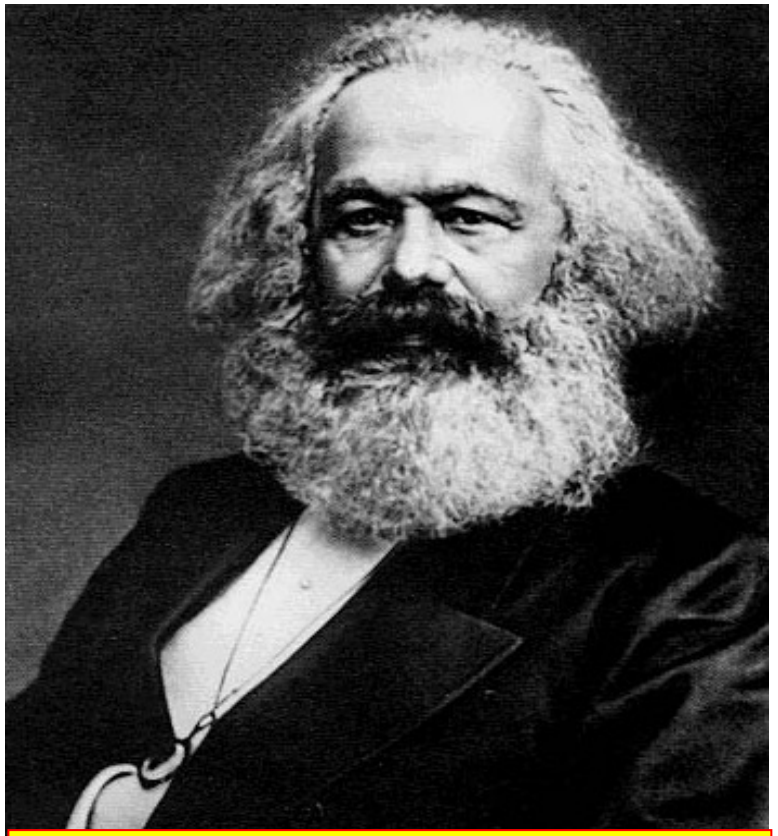
Claude Monet: Camille Monet and child in the garden. (1875)

# A New „Industry Based“ Society.

Black country near Birmingham

New powerful philosophies and political streams developed as a response.

**Karl Marx:**
„The central driving force of capitalism is in the exploitation and alienation of labor”.



Karl Marx, born Trier 1818, died London 1883

General improvement of the economic and social standards, great wealth for a limited group of people, but also serious societal disorders.

”Das Kapital“, published 1867, provided the new political ideology of communism and later socialism.

# Two Large Empires in the Center of Europe.

## A rising nationalism caused massive tensions between European nations.

- **The Austro-Hungarian Empire:**
  - 674.000 km², 53 million people, 17 nations.
  - Severe tensions between Austrians, Hungarians, Czechs, Italians.

The ethnic groups of Austria-Hungary in 1910 according to *Distribution of Races in Austria-Hungary* by William R. Shepherd, 1911.

- **The German Empire (Deutsches Reich) founded 1871.**
- 540.000 km², 64 million people, mainly Germans.
- Massive armament to become a global player.

# Another Large Empire in Eastern Europe.

**An absolute monarchy with a very rich nobility and an extremely poor population: Tsarist Russia moving towards a revolution.**

Marc Chagall, „Witebsk“ (1911).

# At the peak of its evolution Europe decided to go to war.

Albin Egger-Lienz: “Soldaten” (1915).

*Sammlung Schloss Bruck (Lienz)*

# The Decline of the “Master of the Universe”.

**From 1914 till 1945 Europe finally lost everything in 2 huge wars.**

**World War One:**

Cause was mutual distrust between
England, Russia, France, Germany, Austria.

20 million dead

Follow-up: Bolshewik/Stalinistik revolution in Russia.

30 million dead

Global financial crash in 1929, widespread poverty and unemployment in Europe.

Rise of fascism.

**World War Two:**

Cause was aggression by the Nazis (Hitler) supported by Italian fascists (Mussolini).

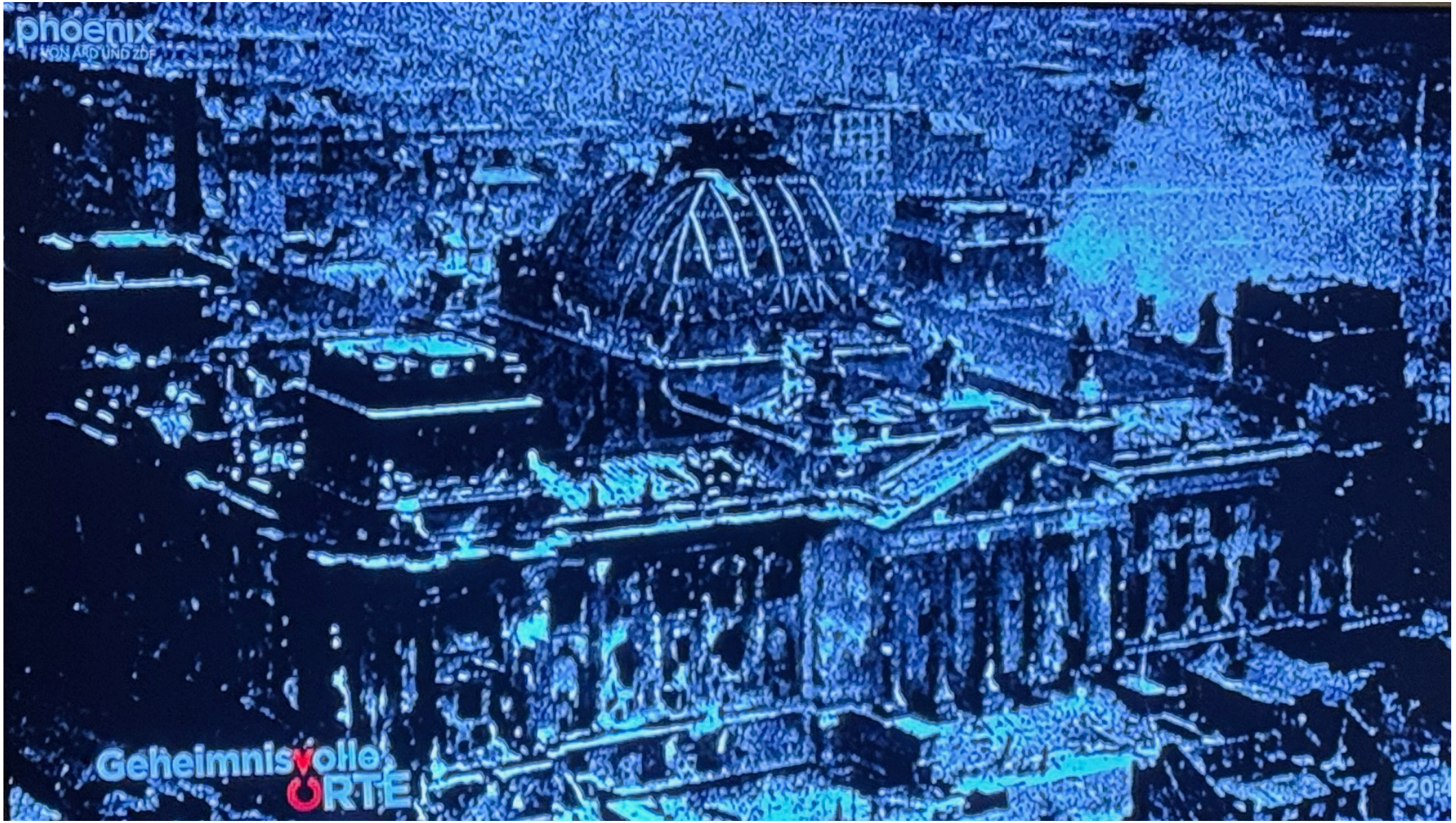
50 million dead

**Central Europe largely destroyed.**
**Millions of refugees.**

The „Master of the Universe“ shrank to a group of small states with little influence.

# The Decline of the “Master of the Universe”.

## Deutscher Reichstag, Berlin 1945.



# The Rise of a New Superpower.

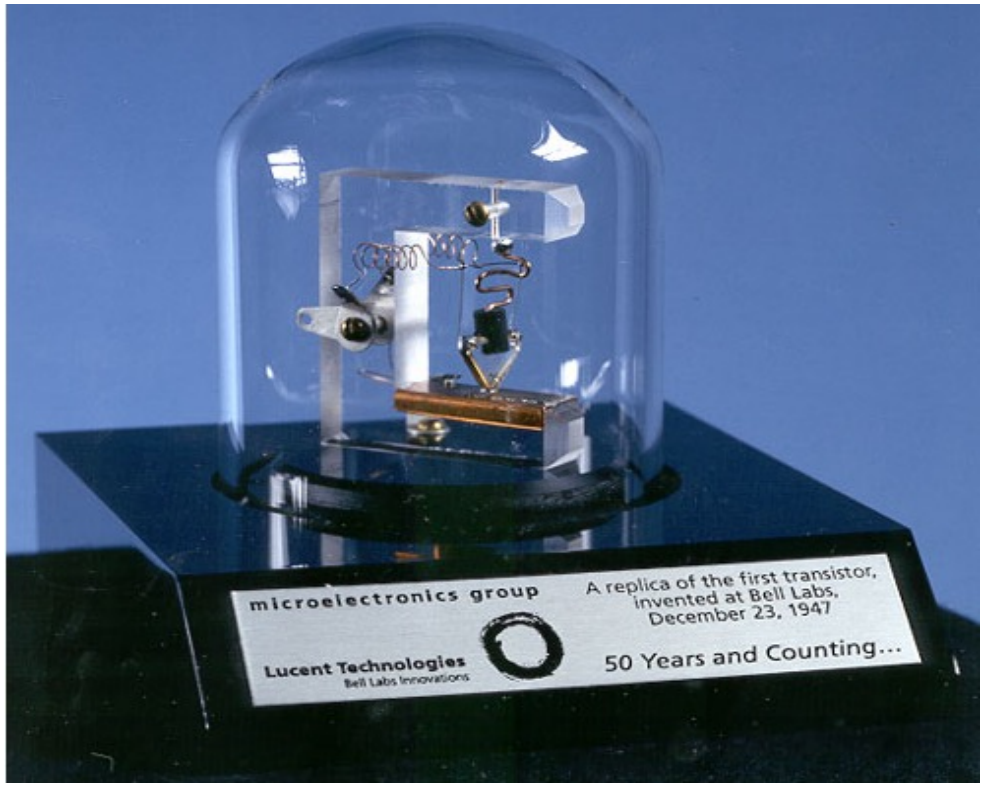
**Iwo Jima, February 23, 1945**
Raising the first flag.

- The USA was the winner in the two World Wars.
- Massive shift of political power from Europe to America.
- USA have become
  - the largest industrial nation in the world,
  - the technological leader,
  - the military superpower,
  - the cultural leader,
  - the global political leader, and
  - the dominating economic and financial power.

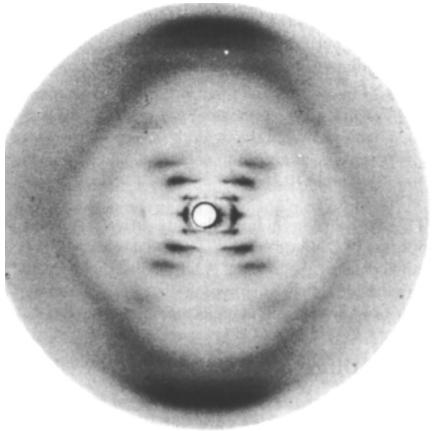
„In my empire the sun never sets.“
(Charles V, King of the Holy Roman-German Empire 1500-1558)

# The Third Industrial Revolution.

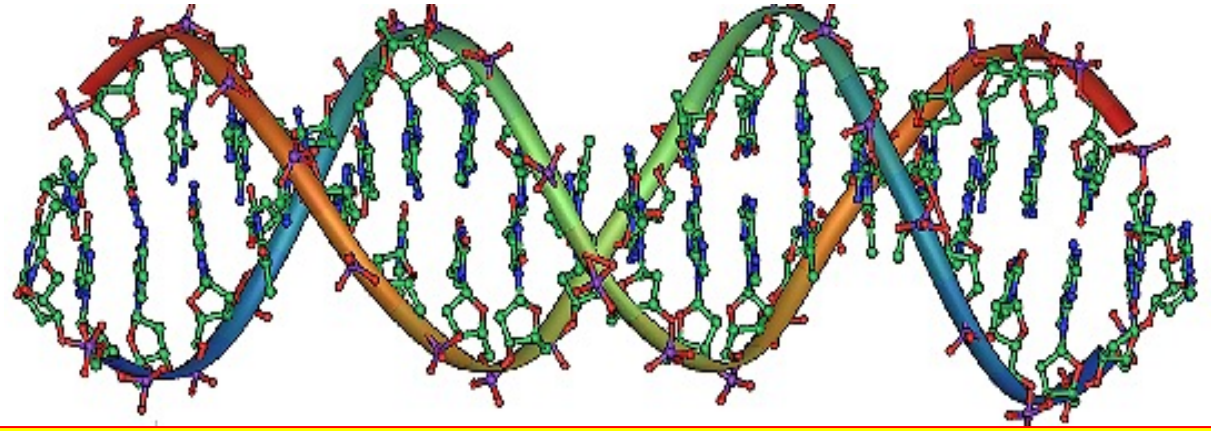
•The second half of the the 20th century is characterised by new revolutionary scientific achievements, like the invention of the transistor, or the clarification of the structure of the DNA, leading to the Third Industrial Revolution.




First transistor (William Shockley, John Bardeen and Walter Brattain at Bell Labs December 23, 1947).



X-Ray diffraction pattern by Rosalind Franklin



Structure of DNA (James Watson, Francis Crick and Rosalind Franklin 1953 at Cambridge, UK)

# A New World: Restructuring and Globalisation.

**•The second half of the 20th century was the period of restructuring:**

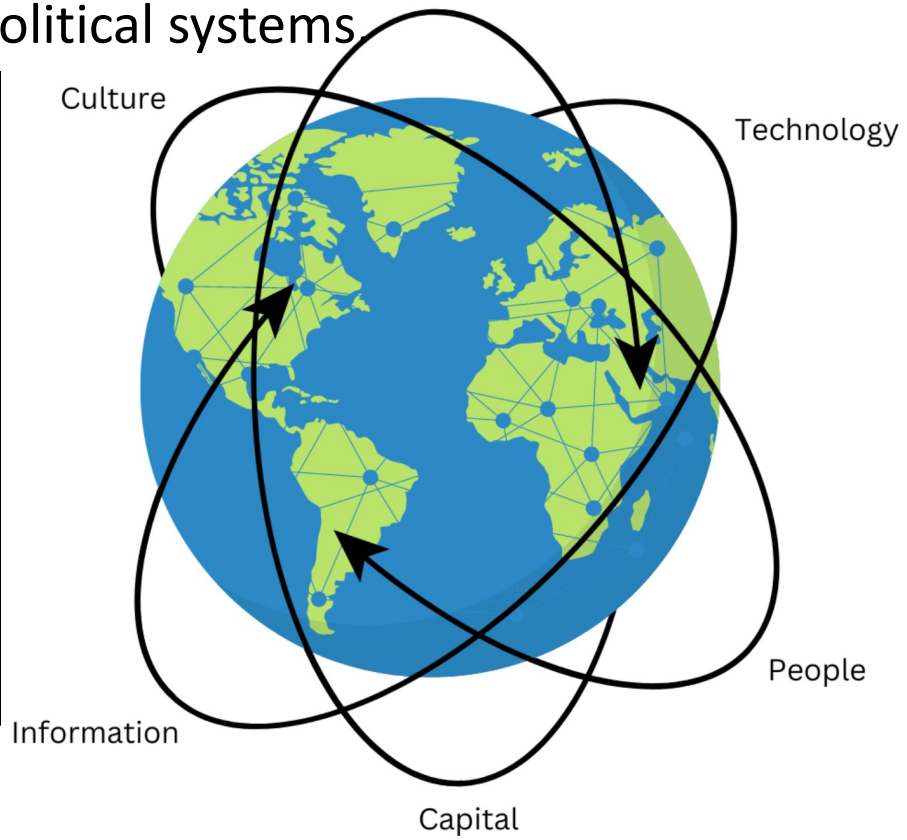
–Rising of the USA to the global superpower.

–Creation of the European Union.

–Emergence of many new states from former colonies.

–Expansion of the democratic political systems.

**Robert Schuman** (1886 - 1963) founder of the European Union.




WTO agreement covers 95% of the world.

**•The 21st century will be the period of globalisation.**

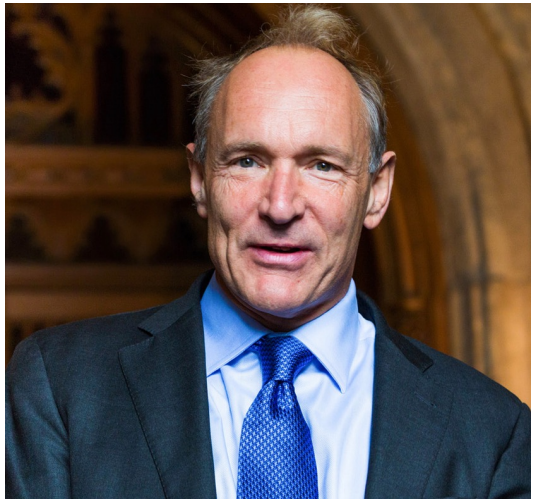
•WTO agreement of 1995 provides for free movement of goods, capital, technology.

•The prime drivers for globalisation were/are

–the USA, Europe and Japan, who were looking for new markets for their industrial and service products, and

–the dynamic developing countries of Asia, like China, India, who provide these markets and act as producers for basic goods.

# The ICT Revolution: Towards the „Global Village“.



- Globalisation means transcending classical geographical boundaries.
- It provides a huge challenge for society.
- ICT has largely impacted on individuals and societies across the globe.

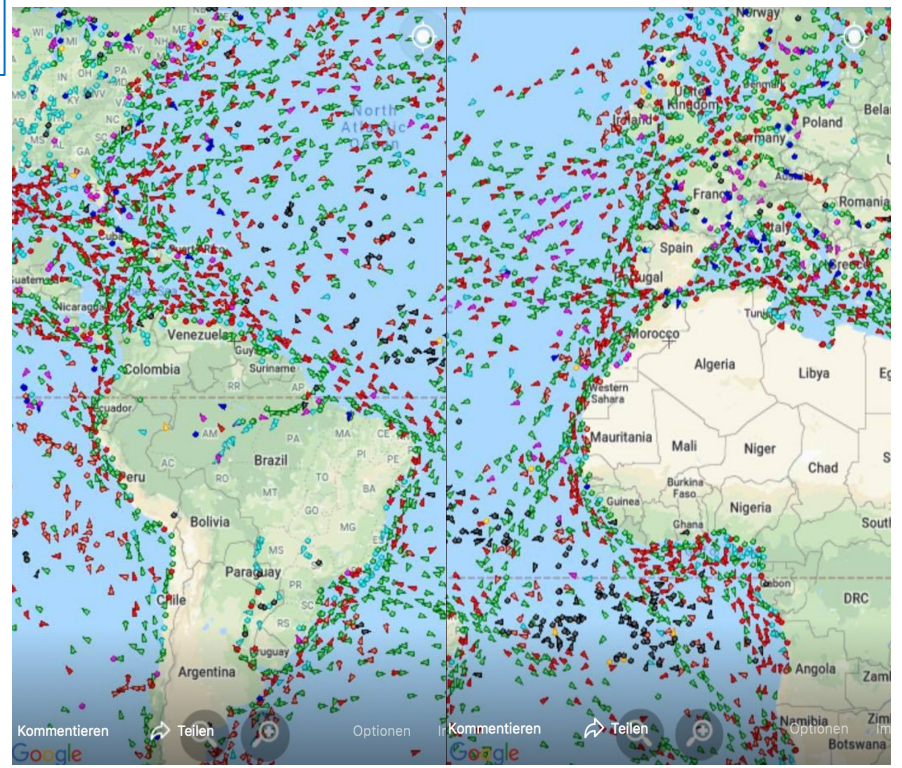
- Tim Berners-Lee, a British scientist at CERN, invented the World Wide Web (WWW) in 1989.
- 6 billion internet users in 2025 (40 million 20 years ago)
- 7 billion smartphones
- 3 billion Facebook users
- Today 8 billion people from 1.000 nations and 200 countries interact, many of them in real-time.

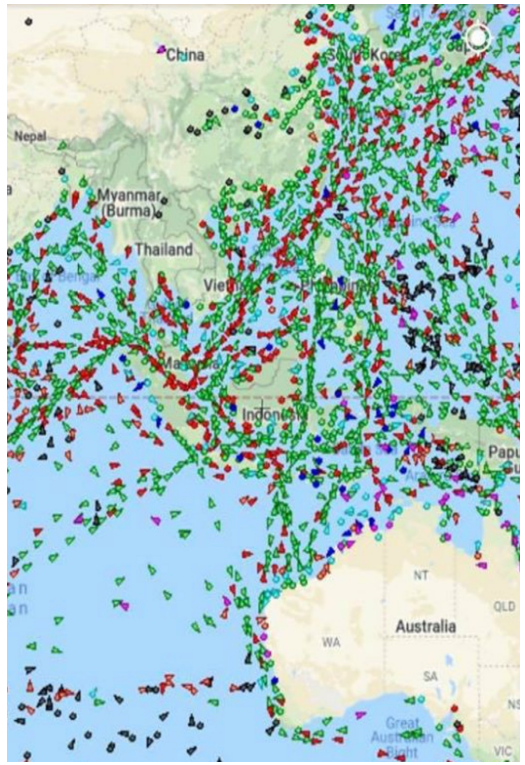
# Global Mobility and Trade.

- **Air traffic:**
  - 26.000 air planes.
  - 40 million flights per year.
  - 10 billion passenger flights (2024).
  - Air traffic likely to double till 2050.

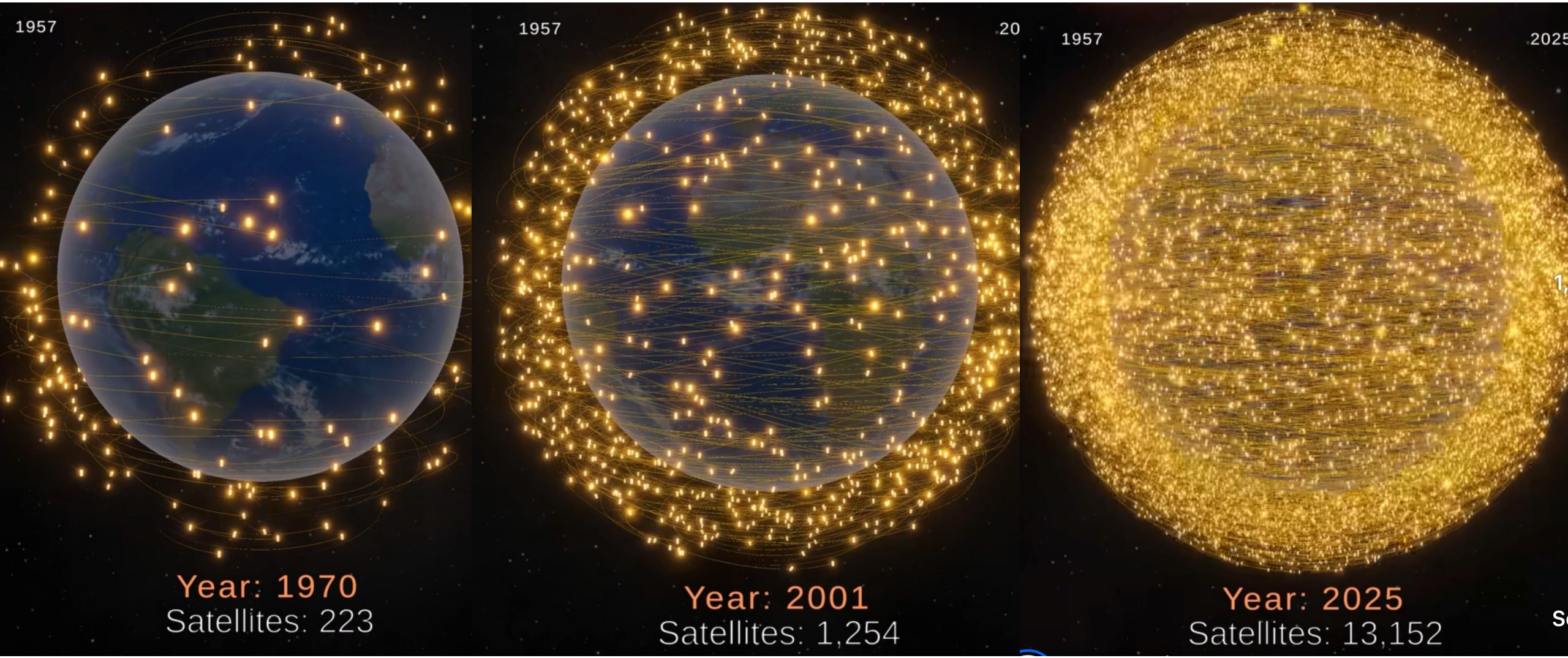
- **Maritime shipping: 90% of global trade of goods.**
  - 60.000 cargo ships (incl. 20.000 oil tankers).
  - 20 million containers in use.
  - 11 billion tons transported annually.
  - Value 25% of global GDP (17.000 billion $).
  - Expected increase 100% by 2050.

Snapshot taken in January 2024:
17.000 planes in the air at the same time.



# Global Communication, Navigation and Earth Observation with Satellites.




Starlink: 10.000 satellites

# Globalization of Industrial Production.

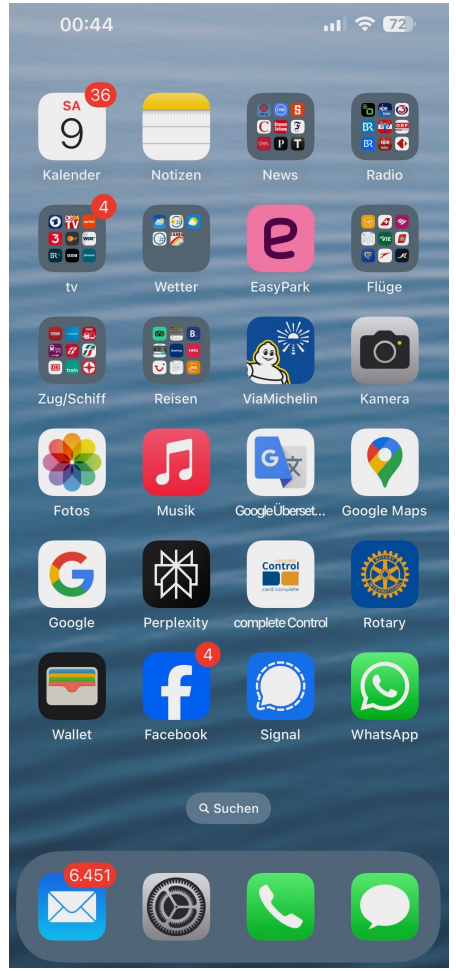
## The iPhone Success Story: "Designed in California, Assembled in China“.

Headquarters Cupertino, CA, USA,  12.000 staff; total staff 160.000.
Gross Profit of Apple 195 billion $, expenditure R&D 40 billion $ (2025).
Value of Apple: 4.000 billion $ (2025)

2007: Nokia market leader 60 million

Apple works with > 700 suppliers
Assembly mainly in Shenzhen and Zhengzhou (China) with 500.000 staff (Foxconn/ Taiwan).

**Since 2007 3 billion iPhones produced.**




2025: Apple market leader 240 million

# Political Globalization.

- Globalization has a major economic and political impact leading to a „multicultural“ global economic and political system – and is irreversible.
- In the 21st century there are four major powers in the world: USA, China, EU and Russia.
- New regional powers are arising around the world (India, Indonesia, Iran, Saudi Arabia, Brasil…..)

**Power constellation in the 21st century.**
*Source: Le Monde Diplomatique 2007*

By 2050 the now dominating triad* US-EU-Japan will be replaced by the triad China-India-USA.

**Triad stands for the group of the 3 leading economic regions.*

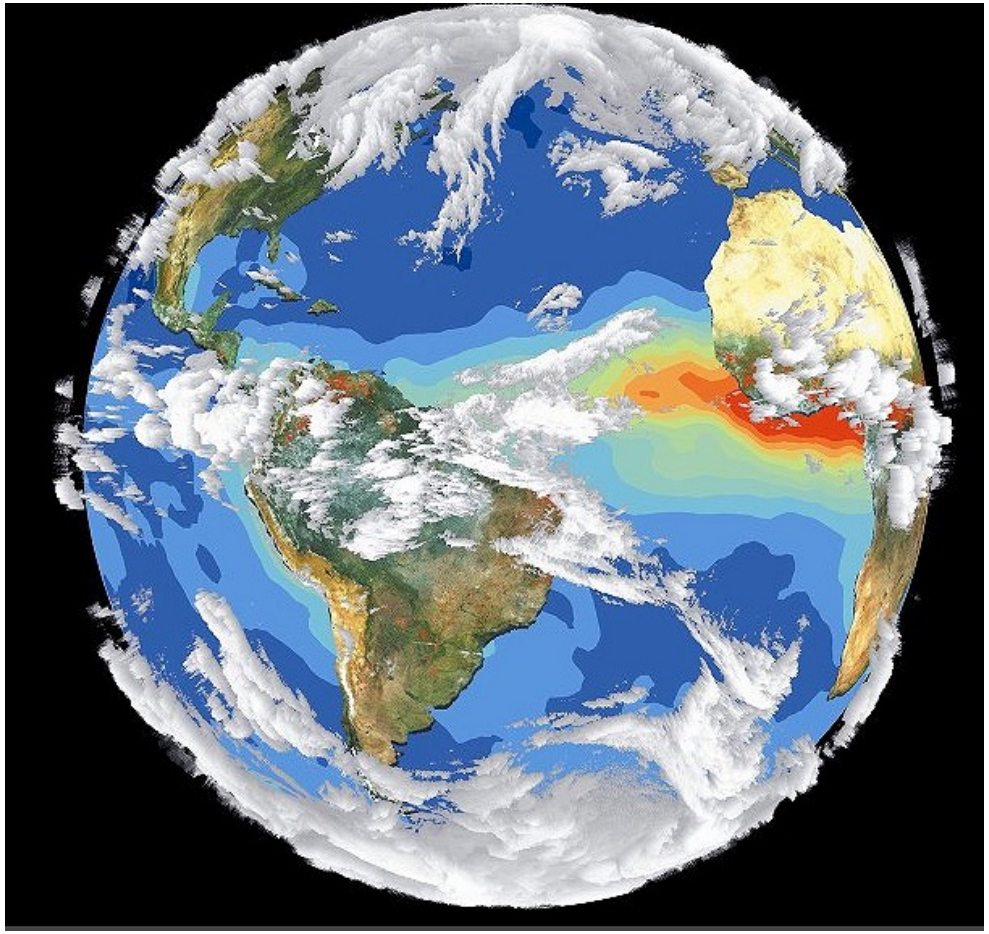
“From the Transatlantic Age to the Pacific Age.”

The unilateral dominant position of the USA will come to an end.

Multipolar world or dual hegemonial system lead by USA/EU/Japan and China/Russia?

# The Anthropocene.

## Human impact on planet Earth:



*Source: NASA, NOAA, ESA-Meteosat, JAXA*

- **Transformation of half of the earth's surface.**
- **10 fold increase in population during last 3 centuries.**
- **15 fold increase in energy and resource consumption during last century.**

Mountaineers climbing the Mount Everest (8.848 m).

*Source: Facebook 2024*

Icon of the Seas:
5.600 passengers